Assignment5

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[5]:
```
# packages
import numpy as np
import matplotlib.pyplot as plt
from matplotlib.ticker import MaxNLocator
from numba import njit              # used for fast compilation (for install␣
↪run "pip install numba")
```

[6]:
```
@njit
# function to perform the Moran Process
# input: population size, number of rounds, payoff matrix, initial number of A␣
↪in the population
# output: frequency array of A
def simulation_moran(N,m, payoff_matrix, initial_A):
    # initialize the population
    pop = np.ones(N,dtype=np.int32)
    pop[:initial_A] = 0             # A -> 0 and B -> 1
    np.random.shuffle(pop)          # shuffle the population

    # track the count of A
    count_A = initial_A
    freq_A_hist = [np.float64(count_A)/N]

    steps = 0


    fitness = np.ones(N, dtype= np.float64) # as basal fitness is 1
    payoff = np.zeros(N, dtype=np.float64)

    # run until fixation (A reaches 0 or N)
    while count_A > 0 and count_A < N:

        # reset payoff and fitness each step
        payoff[:] = 0.0
        fitness[:] = 1.0
```

```
        # calculation of payoffs via random pair-wise interactions
        self_idx = np.random.randint(0,N)    # pick a random number from the␣
↪population for self
        opp_idx = np.random.randint(0,N)     # pick a random number from the␣
↪population for opponent
        while opp_idx == self_idx:           # omit self as opponent
            opp_idx = np.random.randint(0,N)


        # repeat for m round
        for round in range(m):
            # repeat for m round
            payoff[self_idx] += payoff_matrix[pop[self_idx], pop[opp_idx]]
            payoff[opp_idx] += payoff_matrix[pop[opp_idx], pop[self_idx]]

        for k in range(N):
            fitness[k] = 1.0 + payoff[k]

        max_f = np.max(fitness)
        parent_idx = -1
        # selection for reproduction
        while parent_idx == -1:
            candidate = np.random.randint(0, N) # Pick a Random member

            # compare with normalized fitness for reproduction
            if np.random.random() < (fitness[candidate] / max_f):
                parent_idx = candidate # picked for reproduction


        # selection for death
        death_idx = np.random.randint(0, N)

        # exchange
        old_trait = pop[death_idx]
        new_trait = pop[parent_idx]

        pop[death_idx] = new_trait

        # tracking of frequencies
        if old_trait == 0 and new_trait == 1:
            count_A -= 1
        elif old_trait == 1 and new_trait == 0:
            count_A += 1

        steps += 1

        # record frequency of A
```

```
        if steps % N == 0:
            freq_A_hist.append(count_A / N)

    # final frequency add
    if steps % N != 0:
        freq_A_hist.append(count_A / N)

    return freq_A_hist
```

[7]:
```
# parameters
# population size
N = 1000
# number of rounds
m = 4
```

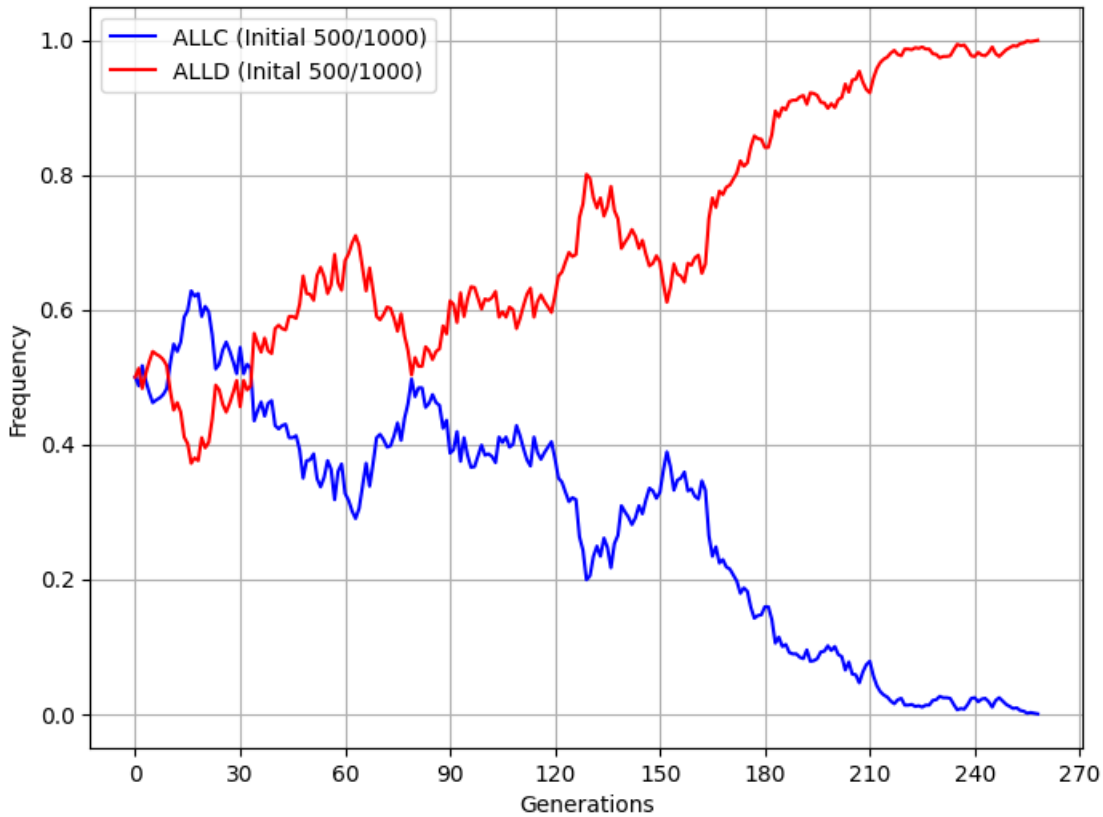
[ ]:
```
# Question 1
# A=ALLC, B=ALLD
# payoff matrix elements
a = 4; b = 1; c = 6; d =2

pd_matrix_1 = np.array([
    [a, b],
    [c, d]
], dtype=np.float64)

print("Simulating game for ALLC vs ALLD...")
freq_ALLC_Q1 = np.array(simulation_moran(N, m, pd_matrix_1, 500))
freq_ALLD_Q1 = 1 - freq_ALLC_Q1
print("Done!")
```

```
Simulating game for ALLC vs ALLD…
Done!
```

[9]:
```
plt.figure(figsize=(8, 6))
plt.plot(freq_ALLC_Q1, label="ALLC (Initial 500/1000)", color='blue')
plt.plot(freq_ALLD_Q1, label="ALLD (Inital 500/1000)", color='red')
plt.xlabel("Generations")
plt.ylabel("Frequency")
plt.gca().xaxis.set_major_locator(MaxNLocator(integer=True))
plt.legend(loc = "best")
plt.grid(True)
plt.show()
```



[12]:
```
# Question 2
# A=TFT, B=ALLD
pd_matrix_2 = np.array([
    [m*a, b + (m-1)*d],
    [c + (m-1)*d, m*d]
], dtype=np.float64)

print("Simulating game for TFT vs ALLD, initial TFT 250)...")
freq_TFT_Q2 = np.array(simulation_moran(N, m, pd_matrix_2, 250))
freq_ALLD_Q2 = 1 - freq_TFT_Q2
print("Done!")

print("Simulating game for TFT vs ALLD, initial TFT 150)...")
freq_TFT_Q3 = np.array(simulation_moran(N, m, pd_matrix_2, 150))
freq_ALLD_Q3 = 1 - freq_TFT_Q3
print("Done!")
```

```
Simulating game for TFT vs ALLD, initial TFT 250)…
Done!
Simulating game for TFT vs ALLD, initial TFT 150)…
Done!
```

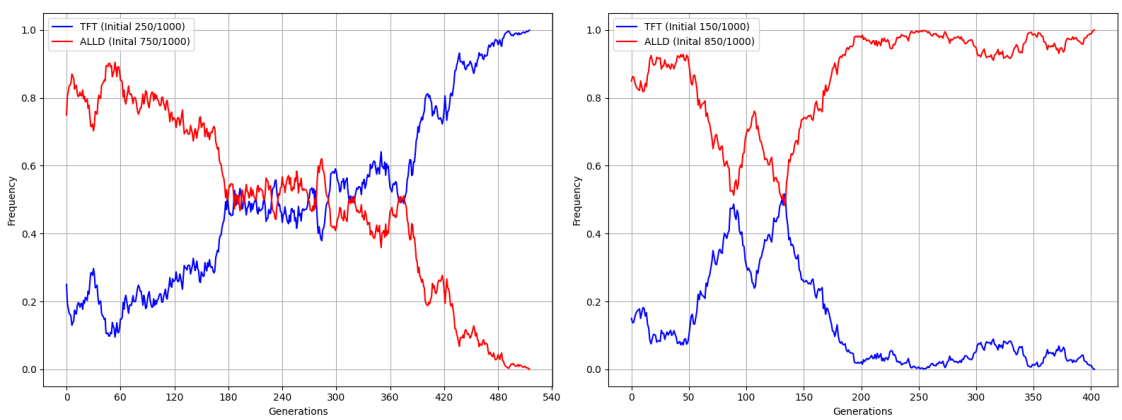
```
Done!
Simulating game for TFT vs ALLD, initial TFT 150)…
Done!
```

[13]:
```
fig, axes = plt.subplots(1, 2, figsize=(16, 6))

axes[0].plot(freq_TFT_Q2, label="TFT (Initial 250/1000)", color='blue')
axes[0].plot(freq_ALLD_Q2, label="ALLD (Inital 750/1000)", color='red')
axes[0].set_xlabel("Generations")
axes[0].set_ylabel("Frequency")
axes[0].xaxis.set_major_locator(MaxNLocator(integer=True))
axes[0].legend(loc = "best")
axes[0].grid(True)

axes[1].plot(freq_TFT_Q3, label="TFT (Initial 150/1000)", color='blue')
axes[1].plot(freq_ALLD_Q3, label="ALLD (Inital 850/1000)", color='red')
axes[1].set_xlabel("Generations")
axes[1].set_ylabel("Frequency")
axes[1].xaxis.set_major_locator(MaxNLocator(integer=True))
axes[1].legend(loc = "best")
axes[1].grid(True)

plt.tight_layout()
plt.show()
```



1 Derivation for Unstable Equilibrium frequency of TFT

From the class notes we can find that the unstable equilibrium point for a PD game can be written as following,

$$\begin{bmatrix} R & S \\ T & P \end{bmatrix}$$

where R is Reward (for mutual cooperation), S is Sucker's payoff, T is Temptation (to defect) and P is Punishment (for mutual defection). Using this notation we can write the unstable equilibrium point is

$$x^* = \frac{P - S}{(R - T) + (P - S)}$$

Now for TFT VS ALLD game we can write the payoff matrix incorporating the number of rounds (i.e. m) as following

	TFT	$ALLD$
TFT	ma	$b + (m-1)d$
$ALLD$	$c + (m-1)d$	md

Now put that R,S,T,P value in equation for x^* and we have the following value

$$x^* = \frac{d - b}{m(a - d) + 2d - b - c}$$

Now put the value of a,b,c,d,m as per the question

$$x^* = \frac{2 - 1}{4 \times (4 - 2) + 2 \times 2 - 1 - 6} = \frac{1}{8 - 3} = \frac{1}{5} = 0.2$$

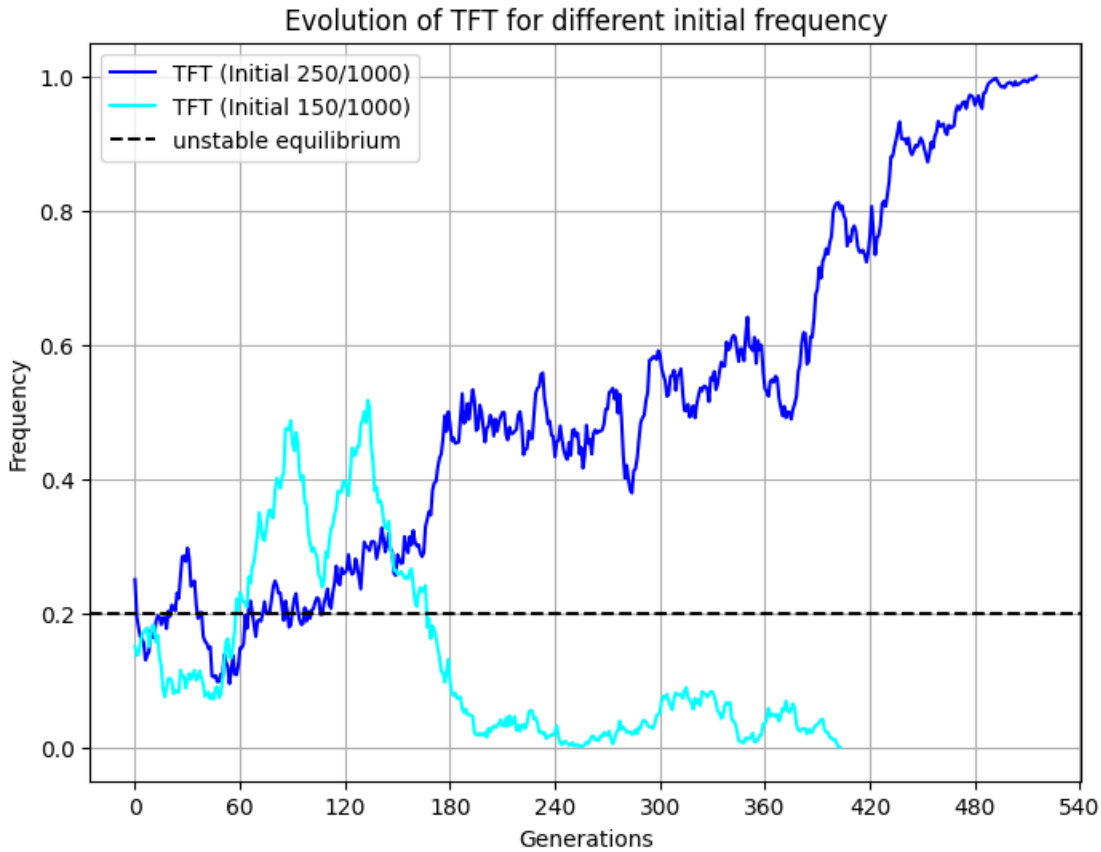
In our case, initial population size is 1000, so $x_{TFT} = 1000 \times (1/5) = 200$ is unstable equilibrium point. Bellow this frequency of TFT, the ALLD should dominate and above this point TFT should dominate the population. And this is convenient with the simulation result!

[14]:
```
plt.figure(figsize=(8, 6))

plt.plot(freq_TFT_Q2, label="TFT (Initial 250/1000)", color='blue')
# plt.plot(freq_ALLD_Q2, label="ALLD (Inital 750/1000)", color='red')
plt.plot(freq_TFT_Q3, label="TFT (Initial 150/1000)", color='cyan')
# plt.plot(freq_ALLD_Q3, label="ALLD (Inital 850/1000)", color='orange')

plt.axhline(y=0.2, color='black', linestyle='--', label='unstable equilibrium')

plt.xlabel("Generations")
plt.ylabel("Frequency")
plt.title("Evolution of TFT for different initial frequency")
plt.gca().xaxis.set_major_locator(MaxNLocator(integer=True))
plt.legend(loc = "best")
plt.grid(True)
plt.show()
```


Evolution of TFT for different initial frequency
TFT (Initial 250/1000)
TFT (Initial 150/1000)
unstable equilibrium
Frequency
Generations
0.0
0.2
0.4
0.6
0.8
1.0
0
60
120
180
240
300
360
420
480
540

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[ ]:
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